
NAME

PseudoHeap

SYNOPSIS

```
use PseudoHeap;

use PseudoHeap qw(:all);
```

DESCRIPTION

PseudoHeap class provides the following methods:

new, AddKeyValuePair, AddKeyValuePairs, DeleteKey, DeleteKeys, DeleteMaxKey, DeleteMinKey, GetCurrentSize, GetKeyType, GetKeyValues, GetKeys, GetMaxKey, GetMaxSize, GetMinKey, GetSortedKeys, GetType, SetKeyType, SetMaxSize, SetType, StringifyPseudoHeap

PseudoHeap is designed to support tracking of a specific number of largest or smallest key/value pairs with numeric or alphanumeric keys along with corresponding scalar or reference values.

Although PseudoHeap is conceptually similar to a heap, it lacks number of key properties of a traditional heap data structure: no concept of root, parent and child nodes; no ordering of keys in any particular order; no specific location greatest or smallest key.

The keys are simply stored in a hash with each key pointing to an array containing specified values. The min/max keys are updated during addition and deletion of key/value pairs; these can be retrieved by accessing corresponding hash.

Addition and deletion of key/value is also straightforward using hashes. However, min/max keys need to be identified which is done using Perl sort function on the keys.

FUNCTIONS

new

```
$NewPseudoHeap = new PseudoHeap(%NamesAndValues);
```

Using specified parameters *NamesAndValues* names and values hash, new method creates a new object and returns a reference to a newly created NewPseudoHeap object. By default, the following property names are initialized:

```
Type = undef;
KeyType = undef;
MaxSize = 10;
```

Examples:

```
$NewPseudoHeap = new PseudoHeap(
    'Type' => 'KeepTopN',
    'KeyType' => 'Numeric');

$NewPseudoHeap = new PseudoHeap(
    'Type' => 'KeepTopN',
    'KeyType' => 'AlphaNumeric',
    'MaxSize' => '20');

$NewPseudoHeap = new PseudoHeap(
    'Type' => 'KeepBottomN',
    'KeyType' => 'AlphaNumeric',
    'MaxSize' => '20');
```

AddKeyValuePair

```
$PseudoHeap->AddKeyValuePair($Key, $Value);
```

Add specified *Key* and *Value* pair to pseudo heap using a new or an existing key and returns PseudoHeap.

AddKeyValuePairs

```
$PseudoHeap->AddKeyValuePairs(@KeyValuePairs);
```

Adds multiple key and value pairs specified in array *KeyValuePairs* to pseudo heap using a new or existing keys and returns PseudoHeap.

DeleteKey

```
$PseudoHeap->DeleteKey($Key);
```

Deletes a specified *Key* from pseudo heap and returns *PseudoHeap*.

DeleteKeys

```
$PseudoHeap->DeleteKeys(@Keys);
```

Deletes a specified *Keys* from pseudo heap and returns *PseudoHeap*.

DeleteMaxKey

```
$PseudoHeap->DeleteMaxKey();
```

Deletes a *MaxKey* along with its associated values from pseudo heap and returns *PseudoHeap*.

DeleteMinKey

```
$PseudoHeap->DeleteMinKey();
```

Deletes a *MinKey* along with its associated values from pseudo heap and returns *PseudoHeap*.

GetCurrentSize

```
$Size = $PseudoHeap->GetCurrentSize();
```

Returns current *Size* of pseudo heap corresponding to number to keys in heap.

GetKeyType

```
$KeyType = $PseudoHeap->GetKeyType();
```

Returns *KeyType* of pseudo heap. Possible *KeyType* values: *Numeric* or *Alphanumeric*.

GetKeyValues

```
@Values = $PseudoHeap->GetKeyValues($Key);  
$NumOfValues = $PseudoHeap->GetKeyValues($Key);
```

Returns an array containing *Values* associated with a specified *Key* in pseudo heap. In scalar context, it returns number of values associated with a key.

GetKeys

```
@Keys = $PseudoHeap->GetKeys();  
$NumOfKeys = $PseudoHeap->GetKeys();
```

Returns an array containing all *Keys* in pseudo heap. In scalar context, it returns total number of keys.

GetMaxKey

```
$MaxKey = $PseudoHeap->GetMaxKey();
```

Returns *MaxKey* present in pseudo heap.

GetMaxSize

```
$MaxSize = $PseudoHeap->GetMaxSize();
```

Returns *MaxSize* of pseudo heap.

GetMinKey

```
$MinKey = $PseudoHeap->GetMinKey();
```

Returns *MinKey* present in pseudo heap.

GetSortedKeys

```
@Keys = $PseudoHeap->GetSortedKeys();  
$NumOfKeys = $PseudoHeap->GetSortedKeys();
```

Returns an array containing all sorted *Keys* in pseudo heap. In scalar context, it retruns total number of keys.

Keys are sorted based on values of *Type* and *KeyType* for pseudo heap:

Type	KeyType	SortOrder	SortOperator
KeepTopN	Numeric	Descending	<=>
KeepTopN	Alphanumeric	Descending	cmp
KeepBottomN	Numeric	Ascending	<=>
KeepBottomN	Alphanumeric	Ascending	cmp

GetType

```
$Type = $PseudoHeap->GetType();
```

Returns *Type* of pseudo heap.

SetKeyType

```
$PseudoHeap->SetKeyType($KeyType);
```

Sets *KeyType* of pseudo heap and returns *PseudoHeap*.

SetMaxSize

```
$PseudoHeap->SetMaxSize($MaxSize);
```

Sets *MaxSize* of pseudo heap and returns *PseudoHeap*.

SetType

```
$PseudoHeap->SetType($Type);
```

Sets *Type* of pseudo heap and returns *PseudoHeap*.

StringifyPseudoHeap

```
$PseudoHeapString = $PseudoHeap->StringifyPseudoHeap();
```

Returns a string containing information about *PseudoHeap* object

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